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1. The Natural Numbers

Exercise 1.1

Prove using mathematical induction that for all positive integers n ,

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

Proof. Let X be the set of all natural numbers for which the formula above holds true. $1 \in X$ because

$$1 = \frac{1(1+1)}{2}.$$

Now assume that $n > 1$, and that $k \in X$ for all $k < n$. We know from our induction hypothesis that the formula holds for $n-1$, so

$$1 + 2 + 3 + \cdots + (n-1) = \frac{(n-1)n}{2}.$$

We add n to both sides to obtain

$$\begin{aligned} 1 + 2 + 3 + \cdots + (n-1) + n &= \frac{(n-1)n}{2} + n \\ &= \frac{n^2 - n + 2n}{2} \\ &= \frac{n^2 + n}{2} \\ &= \frac{n(n+1)}{2}. \end{aligned}$$

Therefore, $n \in X$. It follows from the principle of mathematical induction that $X = \mathbb{N}$. □

Exercise 1.2

Prove using mathematical induction that for all positive integers n ,

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(2n+1)(n+1)}{6}.$$

Proof. Let X be the set of natural numbers from which the formula above holds. $1 \in X$ because

$$1^2 = \frac{1(2 \cdot 1 + 1)(1 + 1)}{6}.$$

Now suppose that $n > 1$, and that $k \in X$ for all $k < n$. In particular, $n-1 \in X$ so

$$1^2 + 2^2 + \cdots + (n-1)^2 = \frac{(n-1)(2(n-1)+1)(n-1+1)}{6}.$$

Adding n^2 to both sides, we deduce that

$$\begin{aligned}
 1^2 + 2^2 + \cdots + (n-1)^2 + n^2 &= \frac{n(n-1)(2n-1)}{6} + n^2 \\
 &= \frac{2n^3 - 3n^2 + n + 6n^2}{6} \\
 &= \frac{2n^3 + 3n^2 + n}{6} \\
 &= \frac{n(2n^2 + 3n + 1)}{6} \\
 &= \frac{n(2n+1)(n+1)}{6}.
 \end{aligned}$$

So $n \in X$, which closes the induction. $\square$

Exercise 1.3

Assuming the triangle inequality ($|x+y| \leq |x|+|y|$) is true, show that $|x_1+x_2+\cdots+x_n| \leq |x_1|+|x_2|+\cdots+|x_n|$ for all $n \in \mathbb{N}$.

Proof. Let X be the set of all natural numbers n such that $|x_1+x_2+\cdots+x_n| \leq |x_1|+|x_2|+\cdots+|x_n|$. $1 \in X$ and it's given that $2 \in X$. Suppose that $k \in X$ for all $k < n$ where $n > 2$. In particular, that means $n-1 \in X$. Therefore, $|x_1+x_2+\cdots+x_{n-1}| \leq |x_1|+|x_2|+\cdots+|x_{n-1}|$. We add $|x_n|$ to both sides to obtain

$$|x_1+x_2+\cdots+x_{n-1}|+|x_n| \leq |x_1|+|x_2|+\cdots+|x_{n-1}|+|x_n|.$$

Let $y = x_1+x_2+\cdots+x_{n-1}$, then the left-hand side of the inequality above becomes $|y|+|x_n|$. Because $2 \in X$, we know $|y+x_n| \leq |y|+|x_n|$. Therefore, $|y+x_n| \leq |x_1|+|x_2|+\cdots+|x_n|$, which closes the induction and shows that $X = \mathbb{N}$. $\square$

Exercise 1.4

Given a positive integer n , recall that $n! = 1 \cdot 2 \cdot 3 \cdots n$ (this is read as n **factorial**). Provide an inductive definition for $n!$. (It is customary to actually use this definition at $n = 0$, setting $0! = 1$.)

Proof.

$$n! = n(n-1)!$$

$\square$

Exercise 1.5

Prove that $2^n < n!$ for all $n \geq 4$.

Proof. Let X be the set of all natural numbers n such that $2^n < n!$. Define $S := X \cup \{1, 2, 3\}$, we show that $S = \mathbb{N}$ by induction. By definition, $1 \in S$. Now suppose that $k \in S$ for all $k < n$ where $n > 1$. If $n = 2$ or $n = 3$, then by definition $n \in S$. If $n = 4$, then $2^4 = 16$ and $4! = 24$. Clearly $16 < 24$, therefore $n \in X$. Now we can solely focus on the cases where $n > 4$. We know $n - 1 \in S$, so $2^{n-1} < (n-1)!$ or $n - 1 \in \{1, 2, 3\}$. But $n > 4$ so $n - 1 \notin \{1, 2, 3\}$. So it must be the case that $2^{n-1} < (n-1)!$. Hence, $2^n < 2(n-1)!$. But $2(n-1)! < n(n-1)! = n!$. Therefore, $2^n < n!$. This closes the induction and shows that $S = \mathbb{N}$. $\square$

Exercise 1.6

Prove that for all positive integers n ,

$$1^3 + 2^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2} \right)^2.$$

Proof. Let X be the set of all natural numbers n such that the equality above holds. We show that $X = \mathbb{N}$ by induction. Because

$$1^3 = \left(\frac{1(1+1)}{2} \right)^2,$$

we know that $1 \in X$. Now suppose that $k \in X$ for all $k < n$ where $n > 1$. In particular, $n - 1 \in X$. Therefore,

$$1^3 + 2^3 + \cdots + (n-1)^3 = \left(\frac{n(n-1)}{2} \right)^2.$$

Adding n^3 to both sides, we have

$$\begin{aligned}
 1^3 + 2^3 + \cdots + (n-1)^3 + n^3 &= \left(\frac{n(n-1)}{2} \right)^2 + n^3 \\
 &= \frac{n^2(n-1)^2 + 4n^3}{4} \\
 &= \frac{n^2(n^2 - 2n + 1) + 4n^3}{4} \\
 &= \frac{n^4 - 2n^3 + n^2 + 4n^3}{4} \\
 &= \frac{n^4 + 2n^3 + n^2}{4} \\
 &= \frac{n^2(n^2 + 2n + 1)}{2^2} \\
 &= \frac{n^2(n+1)^2}{2^2} \\
 &= \left(\frac{n(n+1)}{2} \right)^2.
 \end{aligned}$$

This closes the induction and proves that $X = \mathbb{N}$. □

Exercise 1.7

Prove the familiar **geometric progression** formula. Namely, suppose that a and r are real numbers with $r \neq 1$. Then show that

$$a + ar + ar^2 + \cdots + ar^{n-1} = \frac{a - ar^n}{1 - r}.$$

Proof. Let $S = a + ar + ar^2 + \cdots + ar^{n-1}$.

$$\begin{aligned}
 S &= a + r(a + ar^2 + \cdots + ar^{n-2}) \\
 &= a + r(S - ar^{n-1}) \\
 &= a + rS - ar^n
 \end{aligned}$$

Thus, $(1 - r)S = a - ar^n$ and

$$S = \frac{a - ar^n}{1 - r}.$$

□

Exercise 1.8

Prove that for all positive integers n ,

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}.$$

Proof. Let X be the set of natural numbers n such that the equality above holds. One can quickly verify that $1 \in X$. Now suppose that $k \in X$ for all $k < n$ where $n > 1$. In particular, $n-1 \in X$. Hence,

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n-1)} = \frac{n-1}{n}.$$

Adding $\frac{1}{n(n+1)}$ to both sides and performing some simple algebra, we yield

$$\begin{aligned} \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n+1)} &= \frac{n-1}{n} + \frac{1}{n(n+1)} \\ &= \frac{(n-1)(n+1) + 1}{n(n+1)} \\ &= \frac{n^2 - 1 + 1}{n(n+1)} \\ &= \frac{n^2}{n(n+1)} &= \frac{n}{n+1}. \end{aligned}$$

This closes the induction and shows that $X = \mathbb{N}$. □

Exercise 1.9

By trial and error, try to find the smallest positive integer expressible as $12x + 28y$, where x and y are allowed to be any integers.

Proof. $x = -2$ and $y = 1$ gives 4. □

Exercise 1.10

Use mathematical induction to show that the complete graph of n points has exactly $n(n-1)/2$ lines.

Proof. Let X be the set of natural numbers n for which the formula above works. We show that $S := X \cup \{1\}$ is equal to $\mathbb{N}$ by induction. By definition, $1 \in S$. Suppose that $k \in S$ for all $k < n$ where $n > 1$. Assume $n = 2$, then $2(2-1)/2 = 1$, so $2 \in S$. Assume $n > 2$, we know from our induction hypothesis that $n-1 \in S$, but $n-1 > 1$ and thus cannot be an element of $\{1\}$. So $n-1 \in X$. So the complete graph of $n-1$ has exactly $(n-1)(n-2)/2$ lines. Now add another point to the graph. That point will form a line with every other point (of which there are $n-1$). So

$n - 1$ new lines will be formed. Hence, the complete graph of n will have $(n - 1)(n - 2)/2 + (n - 1)$ lines. Using some algebra,

$$\begin{aligned}\frac{(n - 1)(n - 2) + 2(n - 1)}{2} &= \frac{(n - 1)(n - 2 + 2)}{2} \\ &= \frac{n(n - 1)}{2}.\end{aligned}$$

This closes the induction and shows that $S = \mathbb{N}$, as desired. $\square$

Exercise 1.11

Consider the sequence $\{a_n\}$ defined inductively as follows:

$$a_1 = a_2 = 1, \quad a_{n+2} = 2a_{n+1} - a_n.$$

Use mathematical induction to prove that $a_n = 1$, for all natural numbers n .

Proof. Let X be the set of all natural numbers n such that $a_n = 1$. By the sequence's definition, $1 \in X$ and $2 \in X$. Now suppose that $k \in X$ for all $k < n$ for some $n > 2$. So $a_{n-1} = 1$ and $a_{n-2} = 1$. By definition, $a_n = 2a_{n-1} - a_{n-2} = 2(1) - 1 = 1$. Thereby, $n \in X$ and $X = \mathbb{N}$. $\square$

Exercise 1.12

Consider the sequence $\{a_n\}$ defined inductively as follows:

$$a_1 = 5, \quad a_2 = 7, \quad a_{n+2} = 3a_{n+1} - 2a_n.$$

Use mathematical induction to prove that $a_n = 3 + 2^n$, for all natural numbers n .

Proof. Let X be the set of all $n \in \mathbb{N}$ such that $a_n = 3 + 2^n$. Trivially, $1 \in X$ and $2 \in X$. Now suppose that $k \in X$ for all $k < n$ where $n > 2$. Thus, $a_{n-1} = 3 + 2^{n-1}$ and $a_{n-2} = 3 + 2^{n-2}$. By definition, it follows that $a_n = 3a_{n-1} - 2a_{n-2}$. So $a_n = 3(3 + 2^{n-1}) - 2(3 + 2^{n-2})$.

$$\begin{aligned}a_n &= 3(3 + 2^{n-1}) - 2(3 + 2^{n-2}) \\ &= 9 + 3 \cdot 2^{n-1} - 6 - 2^{n-1} \\ &= 3 + 2^n.\end{aligned}$$

This shows that $n \in X$, and thus $X = \mathbb{N}$. $\square$

Exercise 1.13

Consider the Fibonacci sequence $\{a_n\}$.

(a) Use mathematical induction to prove that

$$a_{n+1}a_{n-1} = (a_n)^2 + (-1)^n.$$

(b) Use mathematical induction to prove that

$$a_n = \frac{(1 + \sqrt{5})^n - (1 - \sqrt{5})^n}{2^n \sqrt{5}}.$$

Proof. (a) Let X be the set of all $n \in \mathbb{N}$ such that $a_{n+1}a_{n-1} = (a_n)^2 + (-1)^n$. We show that $S := X \cup \{1\} = \mathbb{N}$ by induction. By definition, $1 \in S$. Now suppose that $k \in S$ for all $k < n$ where $n > 1$. If $k = 2$, then $a_3a_1 = (2)(1) = 2$ and $(a_2)^2 + (-1)^2 = 1^2 + 1 = 2$. Therefore, $2 \in S$. Now suppose that $n > 2$, then $n-1 \in S$ but $n-1 \neq 1$, therefore $n-1 \in X$. By our induction hypothesis, $a_na_{n-2} = (a_{n-1})^2 + (-1)^{n-1}$.

$$\begin{aligned} a_{n+1}a_{n-1} &= (a_{n-1} + a_n)(a_n - a_{n-2}) \\ &= a_{n-1}a_n - a_{n-1}a_{n-2} + (a_n)^2 - a_na_{n-2} \\ &= a_{n-1}(a_n - a_{n-2}) + (a_n)^2 - ((a_{n-1})^2 + (-1)^{n-1}) \\ &= (a_{n-1})^2 + (a_n)^2 - (a_{n-1})^2 - (-1)^{n-1} \\ &= (a_n)^2 - (-1)^{n-1} \\ &= (a_n)^2 + (-1)^n. \end{aligned}$$

This shows that $n \in X$, which implies $n \in S$, which implies $S = \mathbb{N}$.

(b) First of all, we define

$$\phi := \frac{1 + \sqrt{5}}{2} \quad \text{the golden ratio.}$$

One can confirm using simple algebra that

$$\begin{aligned} 2 - \phi &= (1 - \phi)^2 \\ \phi + 1 &= \phi^2 \\ 1 - \phi &= \frac{1 - \sqrt{5}}{2} \end{aligned}$$

Using the definition and properties of ϕ , we can rewrite the formula as

$$a_n = \frac{\phi^n - (1 - \phi)^n}{\sqrt{5}}$$

Let X be the set of all $n \in \mathbb{N}$ such that

$$a_n = \frac{\phi^n - (1 - \phi)^n}{\sqrt{5}}$$

One can verify that a_1 and a_2 do indeed equal 1 using the formula above, thus $1 \in X$ and $2 \in X$. Now suppose that $k \in X$ for all $k < n$ where $n > 2$. By definition, we know that $a_{n-2} + a_{n-1} = a_n$, so

$$\begin{aligned}
a_n &= a_{n-1} + a_{n-2} \\
&= \frac{\phi^{n-1} - (1-\phi)^{n-1}}{\sqrt{5}} + \frac{\phi^{n-2} - (1-\phi)^{n-2}}{\sqrt{5}} \\
&= \frac{\phi^{n-1} - (1-\phi)^{n-1} + \phi^{n-2} - (1-\phi)^{n-2}}{\sqrt{5}} \\
&= \frac{\phi^{n-2}(\phi + 1) - (1-\phi)^{n-2}(1-\phi + 1)}{\sqrt{5}} \\
&= \frac{\phi^{n-2}(\phi + 1) - (1-\phi)^{n-2}(2-\phi)}{\sqrt{5}} \\
&= \frac{\phi^{n-2}\phi^2 - (1-\phi)^{n-2}(1-\phi)^2}{\sqrt{5}} \\
&= \frac{\phi^n - (1-\phi)^n}{\sqrt{5}}.
\end{aligned}$$

Therefore, we conclude that $n \in X$ and that $X = \mathbb{N}$.

□